

Introduction
to
Programming



Chapter 1 Review

Question:

What are
programs?

```
/**
 * Simple HelloButton() method.
 * @version 1.0
 * @author john doe <doe.j@example.com>
 */
HelloButton()
{
    JButton hello = new JButton( "Hello, wor
    hello.addActionListener( new HelloBtnList

    // use the JFrame type until support for t
    // new component is finished
    JFrame frame = new JFrame( "Hello Button"
    Container pane = frame.getContentPane();
    pane.add( hello );
    frame.pack();
    frame.show();           // display the fra
}
```

Answer:

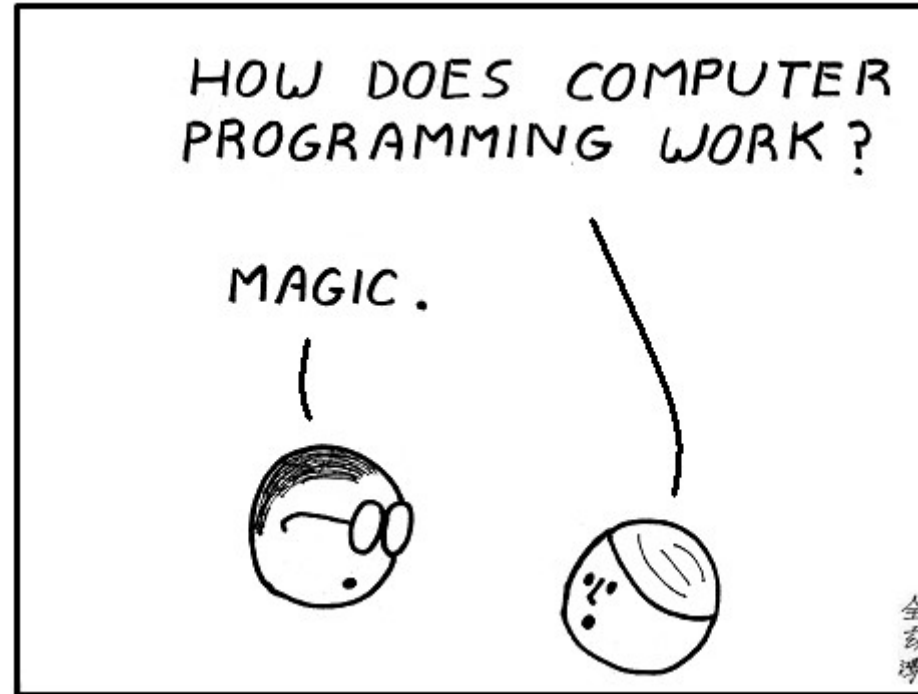
What are
programs?



- Directions or instructions given to a computer.

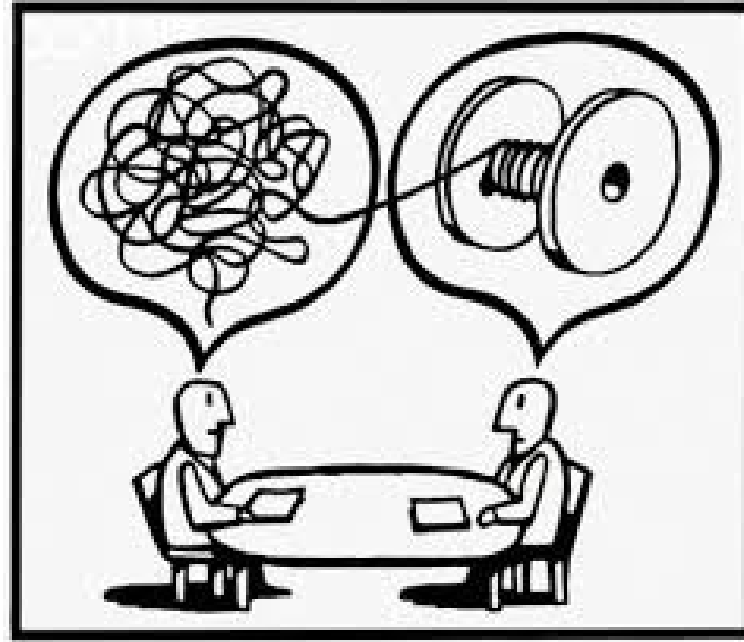
Question:

What is
coding?



Answer:

What is
coding?



- Translating a solution into a language that a computer can understand.

Question:

What is
machine
language?

```
0000000 0000 0001 0001 1010 0010 0001 0004 0128
0000010 0000 0016 0000 0028 0000 0010 0000 0020
0000020 0000 0001 0004 0000 0000 0000 0000 0000
0000030 0000 0000 0000 0010 0000 0000 0000 0204
0000040 0004 8384 0084 c7c8 00c8 4748 0048 e8e9
0000050 00e9 6a69 0069 a8a9 00a9 2828 0028 fdfc
0000060 00fc 1819 0019 9898 0098 d9d8 00d8 5857
0000070 0057 7b7a 007a bab9 00b9 3a3c 003c 8888
0000080 8888 8888 8888 8888 288e be88 8888 8888
0000090 3b83 5788 8888 8888 7667 778e 8828 8888
00000a0 d61f 7abd 8818 8888 467c 585f 8814 8188
00000b0 8b06 e8f7 88aa 8388 8b3b 88f3 88bd e988
00000c0 8a18 880c e841 c988 b328 6871 688e 958b
00000d0 a948 5862 5884 7e81 3788 1ab4 5a84 3eec
00000e0 3d86 dcb8 5cbb 8888 8888 8888 8888 8888
00000f0 8888 8888 8888 8888 8888 8888 8888 0000
0000100 0000 0000 0000 0000 0000 0000 0000 0000
*
0000130 0000 0000 0000 0000 0000 0000 0000
000013e
```

Answer:

What is
machine
language?



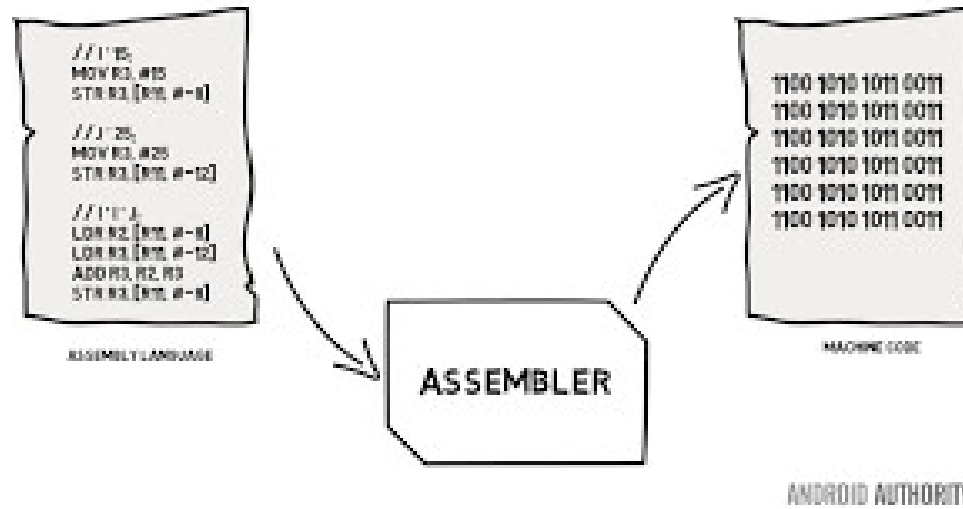
- Instructions written in 0's and 1's.

Question:

What is
Assembly
Language?

```
Line No.
1      .data
2      .align 4
3      A: .long 6, -2, 3, 7, -1
4      .comm min,4,4
5      .comm max,4,4
6      .section .rodata
7      fmt: .string "min = %d, max = %d\n"
8      .text
9      .globl main
10     main:                                     # instruction size
11         pushl %ebp                            # 1 byte
12         movl %esp, %ebp                       # 2 bytes
13         movl A, %eax                          # 5 bytes
14         movl %eax, min                       # 5 bytes
15         movl %eax, max                       # 5 bytes
16         movl $1, %ecx                        # 5 bytes
17     loop:
18         cmpl $5, %ecx                         # 3 bytes
19         jge done                             # 2 bytes
20         movl A(,%ecx,4),%eax                  # 7 bytes
21         cmpl min, %eax                       # 6 bytes
22         jge next1                            # 2 bytes
23         movl %eax, min                       # 5 bytes
24     next1:
25         cmpl max, %eax                       # 6 bytes
26         jle next2                            # 2 bytes
27         movl %eax, max                       # 5 bytes
28     next2:
29         incl %ecx                            # 1 byte
30         jmp loop                             # 2 bytes
31     done:
32         pushl max                            # 6 bytes
33         pushl min                            # 6 bytes
34         pushl $fmt                           # 5 bytes
35         call printf                         # 5 bytes
36         addl $12, %esp                       # 3 bytes
37         movl $0, %eax                       # 5 bytes
38         leave                                # 1 byte
39         ret                                  # 1 byte
```


Answer: What is Assembly Language?



- Uses mnemonics for memory aids. Simplifies programming. Just above machine language.

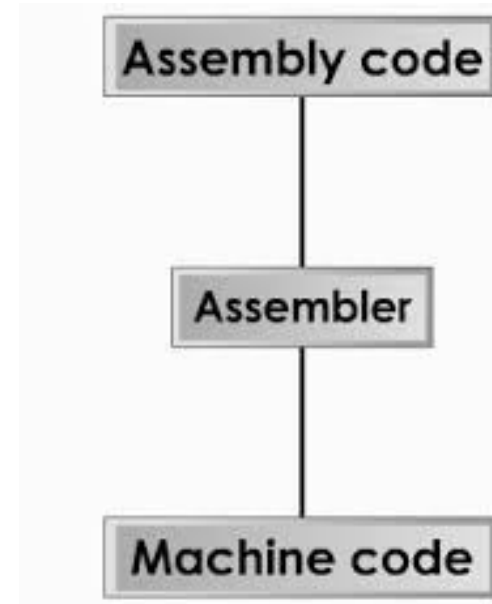
Question:

What is an assembler?



Answer:

What is an assembler?



- A program that converts assembly instructions into machine code.

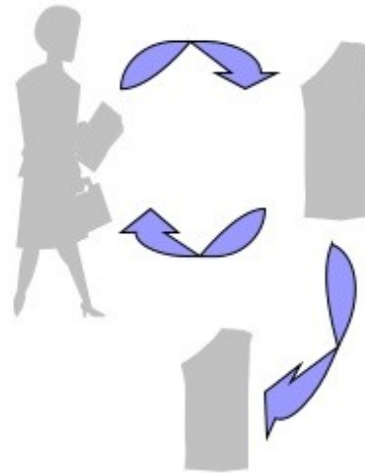
Question:

What is
procedure –
oriented
program?



Procedural vs. Object-Oriented

■ Procedural



Withdraw, deposit, transfer

■ Object Oriented



Customer, money, account

Answer:

What is
procedure –
oriented
program?

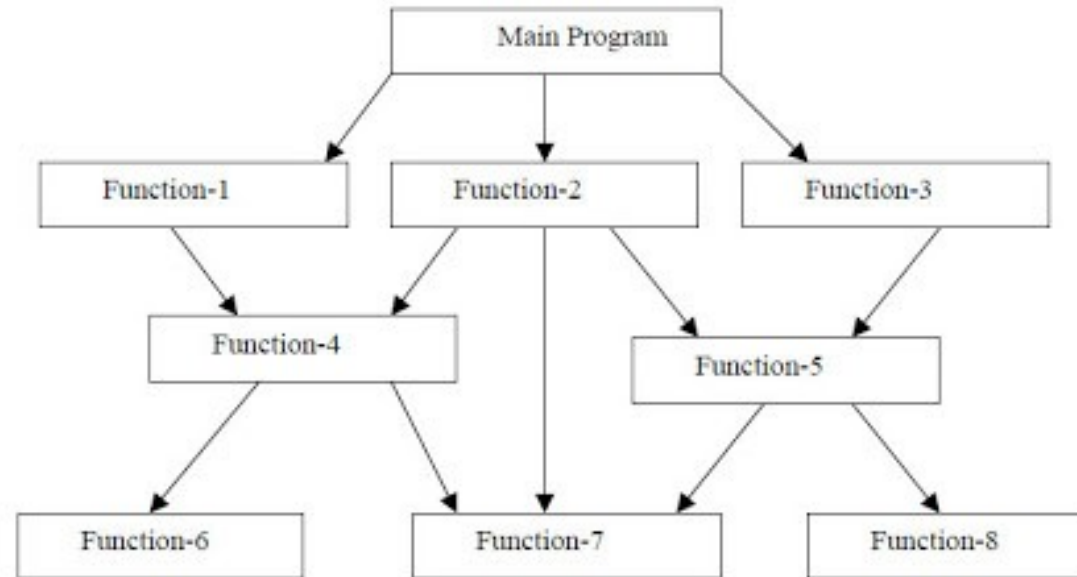


Fig. 1.2 Typical structure of procedural oriented programs

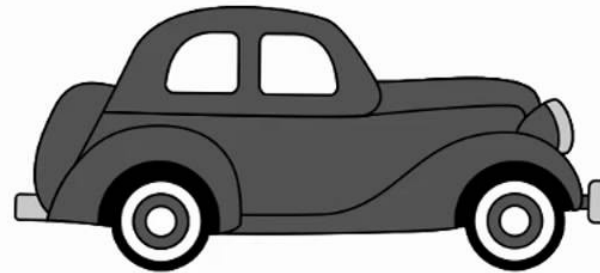
A procedure-oriented program focusses on the procedures or functions that need to occur. For example, payroll, inventory, paychecks.

Question:

What is an
Object
Oriented
program?

OBJECT-ORIENTED PROGRAMMING

Properties
Make
Model
Color
Year
Price



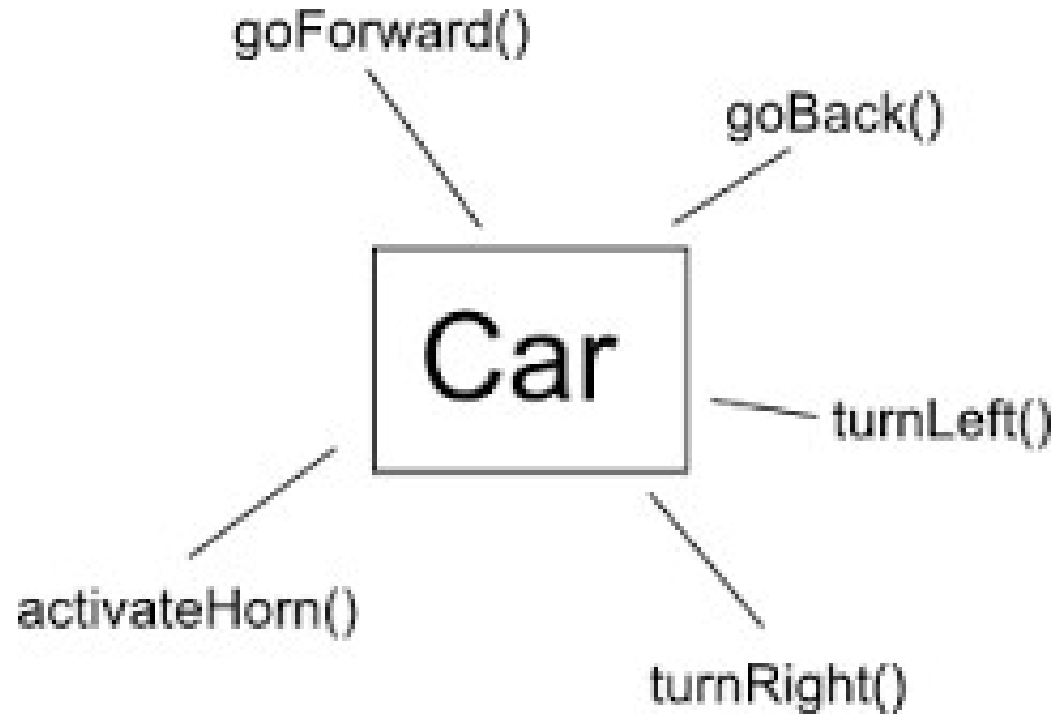
Methods
Start
Drive
Park

Events
On_Start
On_Parked
On_Brake

lynda.com

Question:

What is an
Object
Oriented
program?



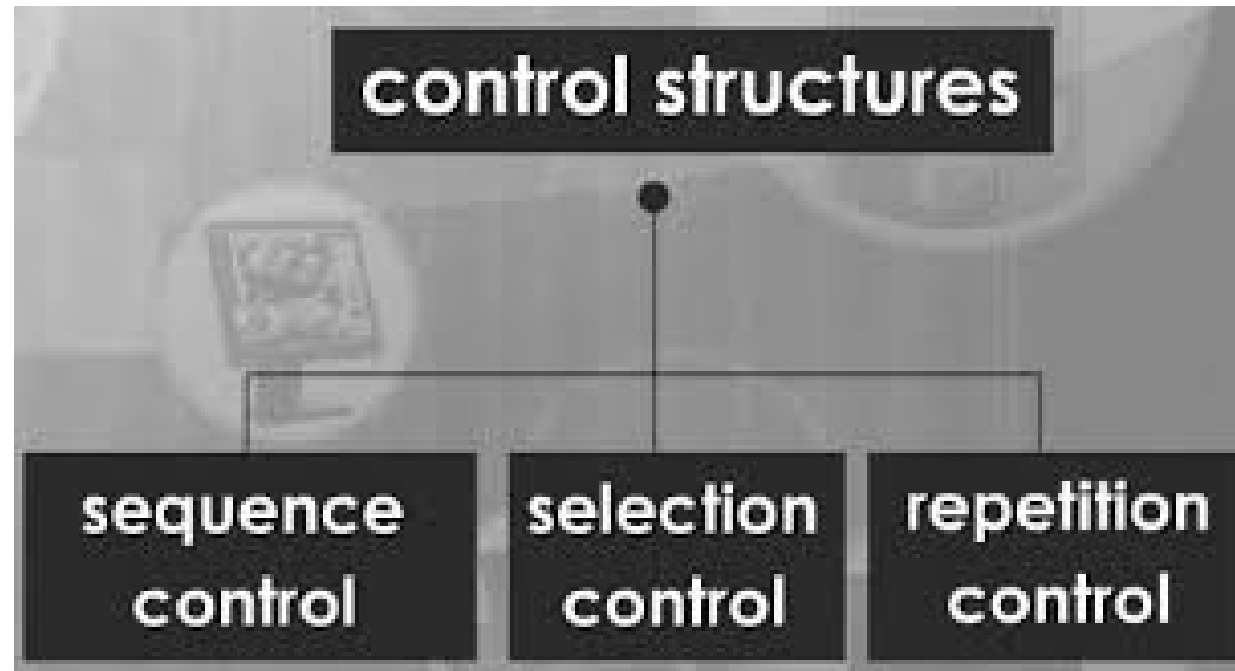
Question:

What are
programming
control
structures?

- Sequence – the order of execution – all programs have it
- Decision - if / else
- Repetition – do while, loop, while

Answer:

What are
programming
control
structures?



Question:
What is an
algorithm?

algorithm

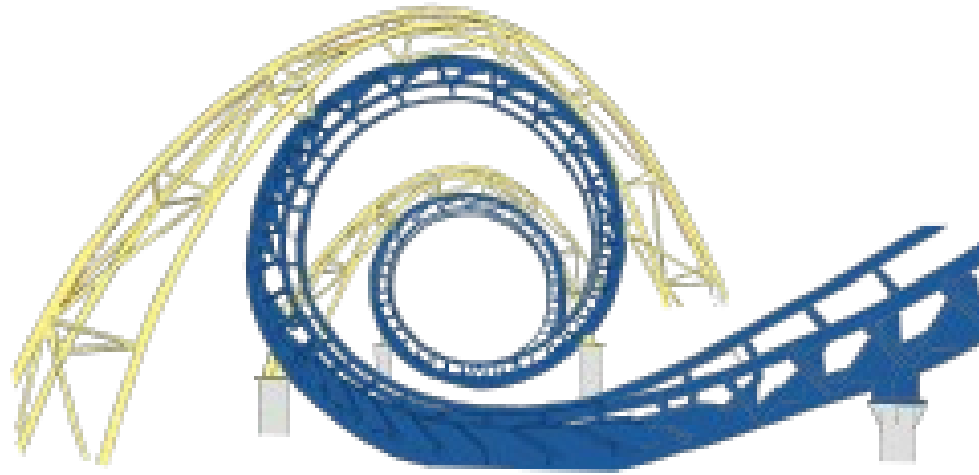
noun

Word used by programmers when
they do not want to explain what
they did.

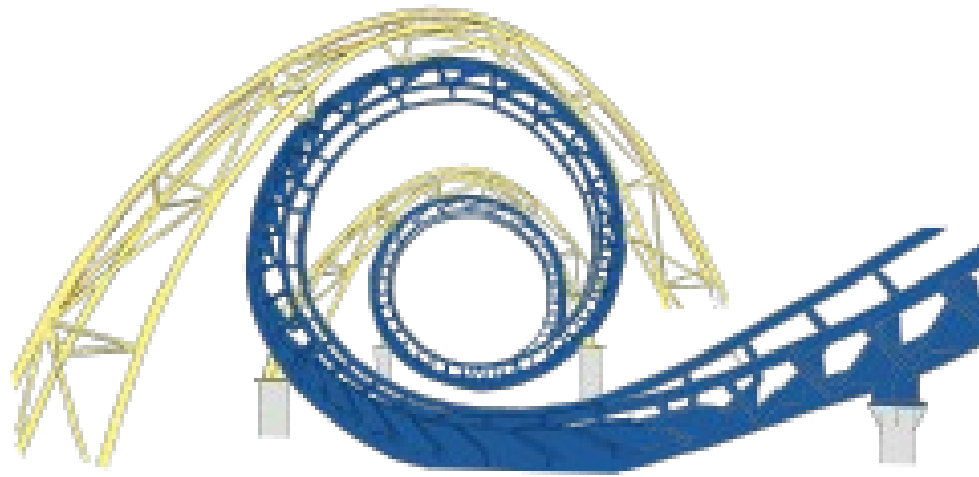
ifunny.co

NOT!!!!

What structure
ends when its
condition is
met?



Answer: What structure ends when its condition is met?



Question:
What control
structure is in
all programs?



Answer:
What control
structure is in
all programs?



Question:
What is a logic
structure?

PREMISE 1

IF $A = B$

PREMISE 2

AND $B = C$

CONCLUSION

THEN $A = C$

Answer: What
is a logic
structure?

- **Same as Control Structures**

Question 1

- Instructions written in 0's and 1's are called ----- language.



Answer 1

- Instructions written in 0's and 1's are called **machine language**.



Question 2

- The three basic control/logic structures are?

Answer 2



- The three basic control/logic structures are:
 - Sequence, Decision/Selection, and Repetition.



Question 3

- Which control structure is in all programs?

Answer 3

- Which control structure is in all programs?
Sequence



Question 4

- The step by step instructions that accomplish a task are called?

Answer 4

The step by step instructions that accomplish a task are called an?

